NICOLÒ CICCARONE

Senior Data Scientist & Applied Statistician | Machine Learning & MLOps

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complexalabs.com

PROFESSIONAL SUMMARY

Senior data scientist and applied statistician with an econometrics background, building predictive models and production machine learning systems. Applies causal inference, survival analysis and time-series forecasting to climate, financial risk and business decisions. Develops data processing applications using Python, SQL, Spark and AWS, with APIs, cloud orchestration and monitoring. Delivers analytics across staffing, retail, aviation and financial services.

TECHNICAL SKILLS

- **Statistical Modeling:** Statistics, econometrics, causal inference, survival and mixture-cure models, score-driven (GAS) models, time-series forecasting, Hawkes processes, climate risk, gradient boosting.
- **Programming & Data:** Python, R, SQL query languages, Polars, PySpark / Apache Spark, Julia (basic), GIS, spatial data, MongoDB; quantitative data management.
- **Machine Learning:** PyTorch, mixture-of-experts (MoE), natural language processing (NLP), predictive models for customer lifetime value (CLTV), churn and demand forecasting.
- **MLOps, Cloud & Serving:** AWS SageMaker, ECS, EKS, ECR, CloudWatch; Docker, Kubernetes, FastAPI, Celery, Grafana; model deployment, production inference, distributed workflows.
- **Additional Platforms:** Databricks; Microsoft Azure ML, Pipelines and Fabric; Google Cloud Platform (GCP) Vertex AI Workbench and Colab Enterprise.

PROFESSIONAL EXPERIENCE

Complexa Labs — Bari, Italy
Quantitative Data Science & Engineering — 01/2025 – Present

- Deliver statistical modeling and production analytics for enterprise clients with over €1B in revenue.
- Built and deployed a time-to-fill forecasting engine on AWS SageMaker using score-driven discrete-time survival models (GAS), right-censored vacancy-day panels and time-varying recruitment funnel data.
- Built customer lifetime value (CLTV) models with mixture-cure survival, gradient-boosted margin regression and calibrated segmentation for revenue forecasting and next-best-product targeting.
- Designed and deployed a macroeconomic scenario simulator to forecast business impacts under changing market conditions, supported by distributed workflow orchestration.
- Optimized distributed data processing pipelines with PySpark joins, window functions and broadcast optimization, **reducing execution time by 70%;** deployed natural language processing (NLP) applications as Python FastAPI microservices.

acmetric B.V. — Amsterdam, Netherlands
Data Scientist — 01/2024 – 11/2024

- Developed and deployed causal artificial intelligence (AI) models for pricing optimization and demand forecasting at Adidas International B.V.; the pricing algorithm remains in production use.
- Developed and maintained gradient boosting machine (GBM) models to predict customer churn and litigation for LATAM Airlines.

MOOS Solutions B.V. Amsterdam, Netherlands
Data Scientist 10/2022 – 12/2023

- Designed and deployed a proprietary mixture-of-experts (MoE) engine for real-time inventory optimization, routing spatial Internet of Things (IoT) sensor data to specialist models; used by a startup. Forecast performance evaluated using mean absolute percentage error (MAPE).

UBS Kraków, Poland
Risk Modeling and Analytics Specialist 04/2022 – 09/2022

- Developed and deployed an initial climate-risk model for a commercial real-estate portfolio, applying scenario-based stress testing aligned with NGFS / IPCC pathways.

EDUCATION & SPECIALIST TRAINING

Università Cattolica del Sacro Cuore 04/2026 – 07/2026
Specialized Training Programme in Statistical Epidemiology
Methods for causal attribution of environmental exposures to cardiovascular, respiratory and immune-mediated public-health outcomes.

Tinbergen Institute, Amsterdam 07/2023
PhD-Level Summer School in Deep Learning and Generative AI
Intensive training with Eran Raviv, PhD.

Vrije Universiteit Amsterdam 2021 – 2022
M.Sc. Econometrics & Operations Research
Grade: 8.0/10. Thesis: *Price volatility and climatic extremes: distinguishing signal from noise.*

University of Bologna 2019 – 2021
M.Sc. Economics, Econometrics Track
Grade: 110/110 cum laude. Thesis: *Hedging against climate change: A Hawkes process approach.*

University of Trento 2015 – 2018
B.Sc. Economics and Management
Grade: 110/110 cum laude. Department of Economics Honours Programme.

Classical High School (Liceo Classico)
Diploma di maturità classica (Classical Studies Diploma)
Grade: 100/100.

LANGUAGES & RECOGNITION

- **Languages:** Italian (native), English (C1/C2), French (B1), German (A2).
- **Recognition:** INPS Merit Scholarship (2019); University of Trento Merit Scholarship (2018).
- **Certification:** TOEFL 107/120 (2021).